

Component	Qty	Voltage	Cont. Current	Stall Current	RPM	Notes
25mm JGA25-370B DC Motor with Encoder	6	6V	0.5A	2.1A	170	25mm fits i wheel (~30mm)
DRV8871 Motor Driver	6	6.5-45V	3.6A max	--	Account for stall current	Learn Adafruit
PCA9658 16-Channel Servo Driver	1	VCC: 3-5V V+: 5-12V	25 mA per pin	--	Connect to power supply to prevent burnout.	Learn Adafruit
ESP32 with 38 pins	1	2.7-3.6V	Average: 80 mA	--	Minimizes pin usage on Raspberry Pi	Robokreed
DS041MG Servos	6	7.4V	150 mA	800 mA	0.12 sec/60°	Torque: 4.0 kg
Zeee LiPo Battery Pack	1	7.4V	Max Current: 260A, Capacity: 5200mAh	Discharge Rate: 50C	Need about 20A for all components peak.	
Raspberry Pi 5		5V	2.5 A	Max: 5A		
LM2596 Buck Converter	2	Input: 3-40V Output: 1.5-35V	2A	Max: 3A		Practice for 4 motors, does create frequency noise
7805 Voltage Regulator	1					To connect to ESP32, does